BOARD OF SECONDARY EDUCATION, KARACHI

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2025

CHEMISTRY

Past Paper 2025

Retyped faithfully from the original board paper — same questions, no changes.

SECTION "A" MULTIPLE CHOICE QUESTIONS (MCQs)

Maximum Marks: 12

Note: NOTE: (i) Attempt all the Questions of this Section. (ii) Do not copy down the part questions. Write only the answer against the proper number of the Question and its part according to the question Paper. (iii) Each question carries 1 Marks

Q1. Choose the correct Answer for each from the given question:

Q1. The unit of active mass is:

(A) gm/cm^3 (B) $mol\ dm^{-4}$

(C) dm^3 (D) mg

Q2. pH value of black coffee is:

(A) 10.2 (B) 12.0

(C) 3.8 (D) 5.0

Q3. It is used for surfacing roads:

(A) Neptha (B) Paraffin

(C) Coal (D) Bitumen

Q4. Protein is derived from Greek word proteios which means:

(A) Second (B) First

(C) Third (D) Zero

Q5. In the reaction $N_2 + 2O_2 \rightleftharpoons 2NO_2$ equilibrium constant expression is:

(A) $|N_2||O_2| / O$ (B) $|NO| / |N_2||O_2|$

(C) $|N_2||O_2| / |NO|^2$ (D) $|NO|^2 / |N_2||O_2|^2$

Q6. The functional group – COOH is used for:

(A) Alkynes
(B) Alcohols
(C) Phenol
(D) Carboxylic Acid

Q7. Benzene is a/an:

(A) Inorganic compound
(B) Heterocyclic compound
(C) cyclic compound
(D) Alicyclic compound

Q8. Glucose is a/an:

(A) Saccharides
(B) Ketoses
(C) Pentosts
(D) aldoses

Q9. This vitamin helps in clotting of blood:

(A) A
(B) D
(C) E
(D) K

Q10. It causes the production of smog:

(A) sulphurdioxide
(B) Carbon monoxide
(C) Nitrogen oxide
(D) Ozone

Q11. Water which does not form lather with soup, is called:

(A) Soft water
(B) Polluted water
(C) Hard water
(D) Distilled water

Q12. Sensitive and technical procedure is:

(A) Advanced Instrumental method
(B) Instrumental method
(C) Spectroscopy method
(D) Classical method

SECTION "B" (Short Answer Question)

Maximum Marks: 24

> Note: NOTE: Answer any EIGHT (8) questions from this section. Each question carries 3 marks.

Q2. Define Organic Chemistry and write two general characteristics of organic compounds.

Q3. Write down three physical properties of water.

Q4. What are carbohydrates? Differentiate between aldoses and ketoses

Q5. Frame structural formulae of the following:

(i) 2.3 dimethyl pentane

(ii) 3 ethyl 1 hexene

(iii) pentyme

Q6. Define

(i) Error

(ii) Precision

(iii) Accuracy

Q7. What are the three spreading factors of pollution? Describe them.

Q8. What is meant by acid rain and how it demolishes monuments?

Q9. What materials are required for preparation of soft drink? Name them.

Q10. Draw flowsheet of soap preparation.

Q11. Find pH, pOH, $[OH^-]$ and $[H^+]$ of 2.46×10^{-9} M KOH solution.

Q12. Write one use of each of the following salts:

(i) Hydrated calcium sulphate

(ii) Iron (II) sulphate hepta-hydrate

(iii) Potassium permanganate.

Q13. Give Reason:

(i) Why number of carbon compounds are more than of silicon compound?

(ii) The higher the altitude of atratosphere the temperature is increased.

Section C (Detailed –Answer Question)

Maximum Marks: 24

> Note: NOTE: Attempt any FOUR (4) questions from this Section. Each question tries (Six) 6 marks.

Q14. State three different theories about concepts of acid and base, with example also indicate one limitation of each concept. Also complete conjugated acid and conjugated base of the following reactions: Acid + Base ⇌ Conjugated acid + Conjugating base | $H_2SO_4 + H_2O$ ⇌ ____ + ____ | HCl + NH_3 ⇌ ____ + ____ | $CH_3COOH + H_2O$ ⇌ ____ + ____

Q15. Elaborate Greenhouse effect (Global Warming)

Q16. Describe any sir water borne diseases.

Q17. What are salts? Write down its preparation by the action of acid on: with representative balanced chemical equation for each:

(i) Metal hydroxide

(ii) Metal carbonate

(iii) Metal oxide

Q18. Define Petroleum. Describe fractional distillation of it. Write name of any five fractions distilled from petroleum with hydrocarbon and show it by the help of labelled diagram.

Q19. What is meant by origin of pharma? Elaborate any five beneficial impacts of pharmaceutical industry in our society.